

Smooth Manifolds and Smooth Maps

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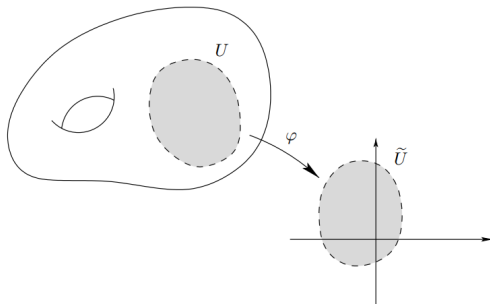
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Topological Manifolds

A topological manifold is a topological space that is

- Hausdorff
- Second countable
- Locally Euclidean
 - A topological space M is locally Euclidean of dimension n if for every point p in M , there exists a **chart** (U, ϕ) where U is open neighborhood of p and $\phi : U \rightarrow \mathbb{R}^n$ is a homeomorphism .

We call ϕ a **coordinate map** or a coordinate system on U



Topological Manifolds

Dimension of topological manifold

- For each point $p \in M$, open subset of $\mathbb{R}^n$ is not homeomorphic to an open subset of $\mathbb{R}^m$.
(*invariance of dimension*) - Prove later
- Constant on each connected component
 $M_n = \{p \in M : \text{local dimension at } p \text{ is } n\}$ is open
 $M = M_0 \sqcup M_1 \sqcup \cdots$
it is impossible for both $C \cap M_n$ and $C \cap M_m$ to be nonempty when $m \neq n$

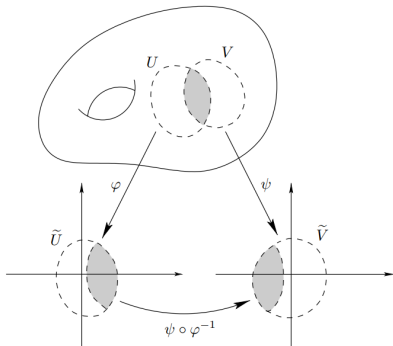
Compatible Charts

Two charts $(U, \phi : U \rightarrow \mathbb{R}^n)$ and $(V, \psi : V \rightarrow \mathbb{R}^n)$ are two charts of a topological manifold are **C^∞ -compatible** if

$$\psi \circ \phi^{-1} : \phi(U \cap V) \rightarrow \psi(U \cap V)$$

$$\phi \circ \psi^{-1} : \psi(U \cap V) \rightarrow \phi(U \cap V)$$

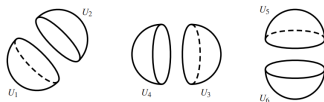
are both C^∞ .



Smooth Atlas

A C^∞ **atlas** on a topological manifold M is a collection $\mathfrak{U} = \{(U_\alpha, \phi_\alpha) : \alpha \in I\}$ of pairwise C^∞ -compatible charts that cover M .

Ex) Charts on a sphere



$$\begin{aligned} U_1 &= \{(x, y, z) \in S^2 \mid x > 0\}, & \phi_1(x, y, z) &= (y, z), \\ U_2 &= \{(x, y, z) \in S^2 \mid x < 0\}, & \phi_2(x, y, z) &= (y, z), \\ U_3 &= \{(x, y, z) \in S^2 \mid y > 0\}, & \phi_3(x, y, z) &= (x, z), \\ U_4 &= \{(x, y, z) \in S^2 \mid y < 0\}, & \phi_4(x, y, z) &= (x, z), \\ U_5 &= \{(x, y, z) \in S^2 \mid z > 0\}, & \phi_5(x, y, z) &= (x, y), \\ U_6 &= \{(x, y, z) \in S^2 \mid z < 0\}, & \phi_6(x, y, z) &= (x, y). \end{aligned}$$

$$\phi_1 \circ \phi_3^{-1}(x, z) = (\sqrt{1 - x^2 - z^2}, z)$$

is C^∞ on $\phi_3(U_1 \cap U_3)$